LECTURE 18

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2 special f^n: Gamma & beta f^n:

Gamma f^n: $\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx \quad n \in R^+$

$\Gamma n = (n-1)\Gamma n-1$, $\Gamma 0 = 1 \quad \Gamma n = (n-1)! \quad n \in N$

Beta f^n: $B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$

$$= \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

Imp: $\int_0^\infty \frac{x^{p-1}}{1+x} = \frac{\pi}{\sin p\pi} = \Gamma p \, \Gamma 1-p \quad 0 < p < 1$

Imp: $\int_0^\infty e^{-y^2} dy = \frac{1}{2}\Gamma\frac{1}{2} \qquad \Gamma\frac{1}{2} = \sqrt{\pi}.$

$B(m,n) = \frac{\Gamma m \, \Gamma n}{\Gamma m+n} \qquad \int_0^{\pi/2} \sin^p\theta \cos^q\theta = \frac{\Gamma\frac{p+1}{2}\Gamma\frac{q+1}{2}}{2\Gamma\frac{p+q+2}{2}} = \frac{1}{2}B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$

$\downarrow$

Wallis's formula.

Proper Integral: $\int_a^b f(x)\,dx \to$ proper if integration is finite & $f(x) \to$ bdd over $[a,b]$

Improper Integral: i) (a,b) infite but $f(x) \to$ bdd over (a,b)

ii) (a,b) finite but $f(x)$ is not bdd over (a,b)

iii) [illegible] $\max f(x) \to$ bdd on (a,b)

ii) (a,b) finite but $f(x)$ is [illegible]

iii) neither (a,b) finite nor $f(x) \to$ bdd on (a,b)

$$\int_1^\infty \frac{dx}{x}, \quad \int_{-\infty}^\infty \frac{dx}{1+x^2}, \quad \int_0^{\pi/2} \cos x \, dx, \quad \int_0^2 \frac{dx}{x(x-1)(x-2)}, \quad \int_0^\infty e^{-x^2} dx$$

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$$\left|\frac{1}{1+x^2}\right| \le 1$$

Improper integral of I st kind : Interval of integration is infinite i.e. either $a = -\infty$ or $b = \infty$ or both & $f(x)$ is bdd.

2nd kind : Interval is finite but $f(x)$ not bdd at 1 or more points in (a,b).

$$\int_0^\infty \frac{1}{1+x^2} \qquad \int_0^3 \frac{dx}{(x-1)(x-2)} \qquad \frac{x^2+1}{x} \quad [1,3]$$

I st kind

$$\int_1^\infty \frac{1}{x^2} dx = \lim_{b\to\infty} \int_1^b \frac{1}{x^2} dx = \lim_{b\to\infty} 1 - \frac{1}{b} = 1$$

$$= -\frac{1}{x}$$

$$\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{\epsilon\to 0} \int_\epsilon^1 \frac{1}{\sqrt{x}} dx = \lim_{\epsilon\to 0} 2(1-\sqrt{\epsilon}) = 2$$

$$\int_0^\infty \frac{1}{1+x^2} dx =$$

$$\int_1^\infty \frac{1}{x^p} dx = ??. \qquad \int_0^1 \frac{1}{x^p} dx = ??.$$